

MACHINE LEARNING BASED WILDFIRE PREDICTION

A Dissertation submitted in partial fulfillment
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by

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BONAFIDE CERTIFICATE

This is to certify that the dissertation entitled “**MACHINE LEARNING BASED WILDFIRE PREDICTION**” submitted by **Ms. PRANATHASREE S** (Reg. No.: **CB.PS.I5DAS22135**), for the award of the **Degree of Master of Science** in Data Science is a bonafide record of the work carried out by her at Department of Mathematics, Amrita School of Physical Sciences, Coimbatore.

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DECLARATION

I, Ms. **PRANATHASREE S** (Register Number - CB.PS.I5DAS22135), hereby declare that this dissertation entitled “**MACHINE LEARNING BASED WILD-FIRE PREDICTION**”, is the record of the original work done by me at Department of Mathematics, Amrita School of Physical Sciences, Coimbatore.

To the best of my knowledge this work has not formed the basis for the award of any degree/diploma/associateship/fellowship/or a similar award to any candidate in any University.

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List of Symbols

X	–	Feature vector of environmental variables
y	–	Target variable (0 = Unburned, 1 = Burned)
$\hat{y}$	–	Predicted output
TP	–	True Positive
TN	–	True Negative
FP	–	False Positive
FN	–	False Negative
Accuracy	=	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	=	$\frac{TP}{TP + FP}$
Recall	=	$\frac{TP}{TP + FN}$
F1 Score	=	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
NDVI	–	Normalized Difference Vegetation Index
NDMI	–	Normalized Difference Moisture Index
NDWI	–	Normalized Difference Water Index
VV	–	SAR Vertical transmit–Vertical receive
VH	–	SAR Vertical transmit–Horizontal receive
TCW	–	Tasseled Cap Wetness
TCG	–	Tasseled Cap Greenness
TCB	–	Tasseled Cap Brightness
SHAP	–	SHapley Additive exPlanations
ROC	–	Receiver Operating Characteristic
AUC	–	Area Under the Curve

Abstract

Wildfires are a major environmental threat to peatland ecosystem because they burn underground and produce enormous amounts of carbon into the atmosphere. It is difficult to predict the occurrence of wildfires in such lands due to the fact that the conventional models depend extensively on the meteorology data and such data is usually inaccessible at the remote locations where there are few ground based monitoring systems. This paper gives a machine learning implemented model of predicting wildfire in Canadian treed peatlands using environmental features, derived through the satellite. Google Earth Engine was used to extract real Sentinel 1 Synthetic Aperture Radar (SAR) and Sentinel 2 multispectral data on the Alberta and Saskatchewan boreal peatland areas between 2019 and 2021. The main environmental characteristics such as vegetation indices (NDVI, NDMI, GNDVI), moisture indicators (Tasseled Cap Wetness, NMDI) and SAR backscatter indices (VV, VH), and six months time slope characteristics were derived and made as inputs to the model. XGBoost algorithm was used to separate the locations of peatlands as burned and non-burned using these features. This model was trained with an 80/20 strata train-test sample and assessed in terms of ROC-AUC, Normalized Gini, Recall, Precision, and F0.5 score. The SHAP (SHapley Additive exPlanations) analysis was utilized to explain the contribution of elements and the most effective ones to determine the most essential predictors of wildfire occurrence. The findings indicate that moisture related indices, especially Tasseled Cap Wetness slope and NDMI, are the most significant predictors of the wildfire possibility, which is in accord with the results published by Pelletier et al. (2023). The model has good discriminative accuracy and gives the spatial wildfire probability maps that would support the early warning systems and decision making regarding fire management in the peatland ecosystems.

Keywords: Wildfire prediction, Machine learning, XGBoost, Remote sensing, Peatland ecosystems, Sentinel 1, Sentinel 2, SHAP, Google Earth Engine.

Chapter 1

Introduction

1.1 Background and Motivation

The development of wildfires has become one of the most serious environmental issues throughout the world. Increased temperatures, extended droughts, along with the altered land use trends have greatly enhanced the intensity and frequency of the events of wildfires [6]. Peatland forests are also among the most sensitive ecosystems that contain huge quantities of organic carbon in their soil [4].

Drying up of peatlands makes it very prone to underground fires, which are hard to detect and put off. Such fires may last long periods and emit high concentration of greenhouse gases, which result in climate change [4].

Conventional wildfire forecasting techniques primarily depend on meteorological information and a ground level monitoring system. These methods are however usually restricted in remote regions where there is low monitoring infrastructure [2]. This demands more scalable and data-driven solutions.

There are new opportunities regarding wildfire prediction because of recent developments in the realms of remote sensing and machine learning technologies. The environmental conditions like the health of the vegetation and moisture level can be captured using satellite data, and complex patterns in this data can be analyzed with machine learning models [7].

1.2 Problem Statement

It is rather difficult to predict the presence of wildfires in peatland ecosystems as the area has not to been continuously monitored, and various environmental factors

including moisture, vegetation stress, and human action are difficult to predict [11].

These associations cannot be well represented by the traditional methods particularly in remote areas. Consequently, an AI based model that can be used to forecast the probability of the wildfires, using environmental characteristics, is necessary.

1.3 Objectives of the Study

The main objective of this project is to develop a machine learning based wildfire prediction system using environmental features inspired by remote sensing data.

The specific objectives are:

- To analyze satellite remote sensing data for the monitoring environmental conditions [10]
- To extract relevant environmental features influencing wildfire occurrence
- To construct time series datasets representing environmental changes [12]
- To implement a machine learning model (XGBoost) for prediction [9]
- To evaluate model performance using standard metrics
- To analyze the effectiveness of feature-based wildfire prediction

1.4 Applications

Machine learning based wildfire prediction systems have several important applications in environmental monitoring and disaster management.

- **Early Warning Systems:** Wildfire prediction models help identify high risk areas in advance, enabling the authorities to issue alerts and take preventive measures before fire events occur [5].
- **Peatland Conservation:** These systems support the protection of peatland ecosystems by detecting conditions that may lead to underground fires and environmental degradation [4].
- **Fire Management Support:** Prediction models assist fire management agencies in resource allocation, improving response time and reducing wildfire damage [5].

- **Infrastructure Risk Assessment:** Wildfire risk maps help identify vulnerable infrastructure such as roads, settlements, and industrial areas located near high-risk zones.
- **Environmental Monitoring:** These models contribute to long term monitoring of environmental changes and help researchers a understand wildfire patterns and climate impacts [13].

1.5 Literature Review

Several studies have explored the use of machine learning and remote sensing techniques for wildfire prediction. These approaches focus on improving fire risk assessment using environmental and satellite based data.

1.5.1 Machine Learning for Wildfire Prediction

Machine learning based wildfire prediction models have been highly used in that they can train intricate trends on the basis of historical data. These models are applied to analyse different environmental factors such as temperature, humidity, vegetation indices and past fire incidences so as to project the risk of the wild fire.

The Gradient Boosting can handle large data sets and nonlinear relationships, and are typically utilized frequently due to the fact that offer capability [8].The Gradient Boosting are capable of large data sets and nonlinear relationships and are typically used frequently because of the fact that they offer such capability.One of them is XGBoost which is popular due to high levels of accuracy and efficiency of the tool.

It is especially the machine learning models that are successful as they are able to potentially introduce different types of data, including satellite readings and environmental variables, in more predictive and evidence based predictions compared to the traditional models.

1.5.2 Remote Sensing of Peatland Moisture and Fire Risk

The application of remote sensing is a huge task of monitoring the environmental conditions in extensive geographical areas. Satellite Imaging provides the real time information about the condition of vegetations, temperatures on the surface, and the water content.

As a rule, to control the vegetation state and moisture content, such indices as NDVI and NDMI are commonly used to evaluate a state of plants and the environment [10]. The time when such indices are low is an indication that the vegetation is dry and stressed, and it increases the likelihood of wildfires happening.

In addition, the Synthetic Aperture Radar (SAR) can be used to record surface moisture data in cloudy weather conditions as well [3]. That leaves SAR particularly relevant to peatland ecosystems where moisture content also is the primary determinant of fire vulnerability. The dry peatlands are very vulnerable to underground fires and as such the moisture should be monitored in the prediction of the wild fire.

1.5.3 Seasonally Detrended Time Series for Drought Characterisation

Time-series analysis involves time change analysis of the environmental conditions. Seasonal trend decomposition is one of these methods and it breaks a time series down into trend, seasonal and residual components [12] of such methods is seasonal trend decomposition that decomposes a time series into trend, season and residual components.

The trend component represents the long term variations like the gradual drying and the seasonal component represents the periodic variations. The residual value has the short term fluctuations.

The strategy facilitates detection of unnatural environmental trends including the long dry period that could elevate fire risks in the wild. Consideration of those patterns will help to investigate drought conditions and enhance the models of prediction of wildfires.

1.5.4 Research Gap

Even though the previous research has produced encouraging findings, most of the methods imply the intricate processing of data and the use of the most advanced satellite methods. This complicates the implementation of this in real life situations.

The work is based on the methodology described in the article in [1], and it implements a simplified machine learning model based on environmental factors to showcase the process of wildfire prediction.

1.6 Organisation of the Report

This report is organized into four main chapters. Chapter 1 introduces the problem, objectives, and background of wildfire prediction. Chapter 2 presents the proposed methodology, including dataset preparation, feature extraction, and model development. Chapter 3 discusses the results and evaluation of the model, and Chapter 4 concludes the study and outlines possible directions for future work.

Chapter 2

Methodology

This chapter describes the methodology adopted for predicting wildfire likelihood in peatland ecosystems using machine learning techniques. The proposed approach is inspired by the satellite based environmental monitoring and integrates feature engineering, model training, and probability estimation.

2.1 Overview

The general process includes preparation of data, extraction of features, model training and prediction. A machine learning model is trained using environmental characteristics based on the remote sensing concepts to classify the risk of wildfires. The model expects to model a real world system of wildfire prediction in a simplified and computationally optimal way..

2.2 Mathematical Prerequisites

2.2.1 Gradient Boosting and XGBoost

Gradient Boosting is an ensemble learning technique where multiple decision trees are combined sequentially to improve prediction accuracy. Each new tree corrects the errors of the previous trees.

XGBoost is an optimized implementation of gradient boosting designed for efficiency and performance. It builds trees in a sequential manner and minimizes a loss function using gradient descent.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where f_k represents the decision trees.

2.2.2 Evaluation Metrics

To evaluate the performance of the wildfire prediction model, the following metrics are used:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics provide a comprehensive understanding of model performance.

2.3 Dataset

The dataset used in this study is generated using Google Earth Engine (GEE) by extracting satellite based environmental features from the peatland regions.

2.3.1 Data Sources

The dataset is constructed using multiple satellite data sources:

- **Sentinel-2 Multispectral Data:** Used to compute vegetation and moisture indices such as NDVI, NDMI, NDWI, EVI, and Tasseled Cap features (Wetness, Greenness, Brightness).
- **Sentinel-1 SAR Data:** Provides VV and VH backscatter values, which are useful for capturing surface moisture conditions under all weather conditions.
- **MODIS Burned Area Data:** Used to generate fire labels. Burned regions are assigned as value of 1, and non-burned regions are assigned 0.

2.3.2 Feature Description

Each sample in the dataset represents a geographical location within the study area. For each location, the following features are extracted:

- Vegetation indices (NDVI, NDMI, NDWI, EVI)
- Moisture and drought indicators (NMDI, Tasseled Cap features)
- SAR backscatter values (VV, VH)
- Temporal features such as slope of NDVI, NDMI, and TCW between two time periods

These features capture environmental conditions that influence wildfire occurrence, such as vegetation health, moisture levels, and drying trends.

2.3.3 Sample Generation

Random sample points are generated within the study region using Google Earth Engine. Satellite derived features and fire labels are extracted for each sample point.

2.3.4 Dataset Composition

The dataset consists of approximately 1000 samples, balanced between fire and non-fire classes. This balanced dataset helps improve the performance and stability of the machine learning model.

2.3.5 Dataset Format

The final dataset is exported as a CSV file, where each row represents a sample location and each column represents a feature or the corresponding fire label.

2.4 Data Preprocessing

After extracting the dataset using Google Earth Engine, basic preprocessing steps are applied before training the machine learning model.

The dataset is first cleaned by selecting only relevant feature columns and removing unnecessary attributes. The input features (X) and target variable (y) are defined, where the target represents wildfire occurrence.

Any missing or invalid values are handled appropriately to ensure data quality. Since XGBoost is a tree based model, feature scaling or normalization is not required.

These preprocessing steps ensure that the dataset is properly structured and ready for model training.

2.5 Feature Selection

Feature selection is performed to identify the most relevant variables for wildfire prediction.

The selected features include:

- Vegetation indices such as NDVI, NDMI, NDWI, and EVI
- Moisture-related indicators such as Tasseled Cap Wetness (TCW)
- SAR backscatter values (VV and VH)
- Temporal features such as slope of NDVI, NDMI, and TCW

These features are chosen based on their ability to capture vegetation health, moisture conditions, and environmental changes over time.

By selecting relevant features, the model is able to learn meaningful patterns and improve prediction accuracy.

2.6 Model Training

The wildfire prediction problem is formulated as a binary classification task, where the objective is to predict whether a wildfire will occur based on environmental features.

2.6.1 Train–Test Split

The dataset is divided into training and testing sets to evaluate model performance. Approximately 80% of the data is used for training, while the remaining 20% is used for testing.

2.6.2 XGBoost Classifier

The XGBoost (Extreme Gradient Boosting) algorithm is used as the primary machine learning model. XGBoost is an ensemble learning technique that builds multiple decision trees sequentially to improve prediction accuracy.

It is chosen due to its ability to handle nonlinear relationships, robustness to different feature types, and strong performance in classification tasks.

2.6.3 Training Process

During training, the model learns patterns from the input features such as vegetation indices, SAR backscatter values, and temporal trends. Each decision tree in the model focuses on correcting the errors of the previous trees.

The trained model is then used to predict wildfire occurrence on the test dataset.

2.6.4 Prediction

The model outputs a binary prediction:

- 1 indicates wildfire occurrence
- 0 indicates no wildfire

These predictions are compared with actual labels to evaluate model performance.

2.7 Visualisation Methodology

Visualization techniques are used to interpret the model predictions and understand the contribution of different environmental features to wildfire occurrence.

2.7.1 SHAP Feature Importance Analysis

SHAP (SHapley Additive exPlanations) is used to interpret the output of the XGBoost model by quantifying the contribution of each feature to the prediction. SHAP values are based on game theory and assign an importance score to each feature for a given prediction.

A SHAP summary plot is generated to visualize how different features influence the wildfire prediction. Each point in the plot represents a sample, and the color

indicates the magnitude of the feature value. Features with higher SHAP values have a stronger impact on increasing wildfire likelihood.

The analysis shows that vegetation indices such as NDVI and moisture related features such as NDMI play a significant role in wildfire prediction. SAR backscatter values (VV and VH) also contribute to capturing moisture variations in peatland regions.

2.7.2 Feature Importance Ranking

The mean absolute SHAP values are computed for all features to rank their importance. Features with higher mean SHAP values are considered more influential in predicting wildfire risk.

The results indicate that NDVI, SAR backscatter (VV, VH), and moisture related indices are among the most of important predictors of wildfire occurrence.

2.7.3 Prediction Interpretation

SHAP analysis assists in making sense of the effects of varying conditions of the environment on the possibility of wildfires. The positive SHAP values mean that the probability of wildfire increases whereas negative values mean that the probability decreases. This gives the machine learning model enhanced interpretability and transparency.

Chapter 3

Results and Discussion

This chapter presents the results obtained from the implementation of the wildfire prediction model using satellite derived environmental features. The analysis is structured to first understand the data characteristics, followed by model behavior, and finally the predictive performance.

3.1 Feature Distribution Analysis

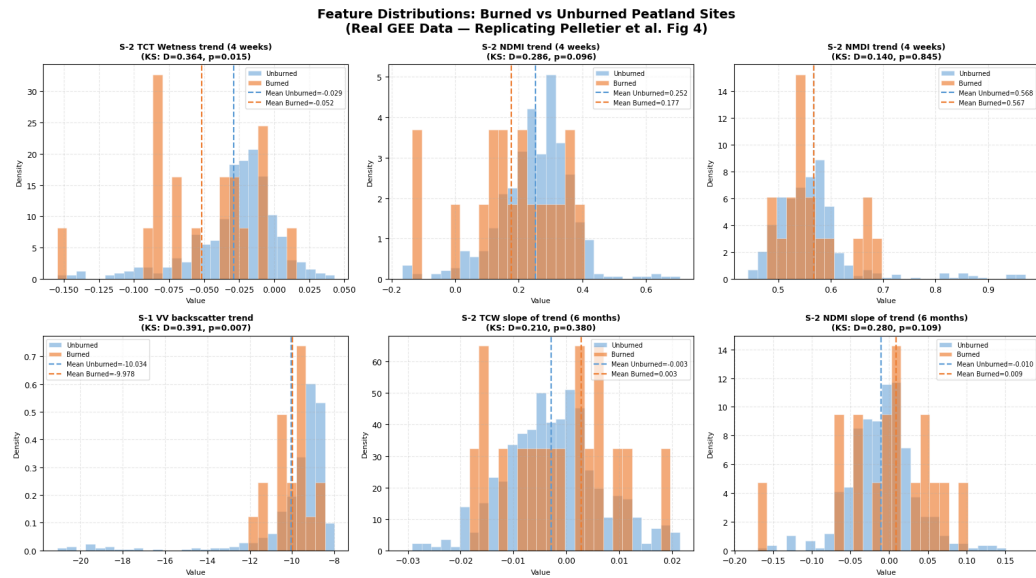


Figure 3.1: Distribution of environmental features for burned and unburned samples

The distribution plots show clear differences between burned and unburned regions. Burned samples generally exhibit lower vegetation indices and reduced moisture levels,

indicating environmental stress conditions prior to wildfire events. This confirms that vegetation dryness of and moisture depletion are strong indicators of wildfire occurrence.

3.2 Correlation Analysis

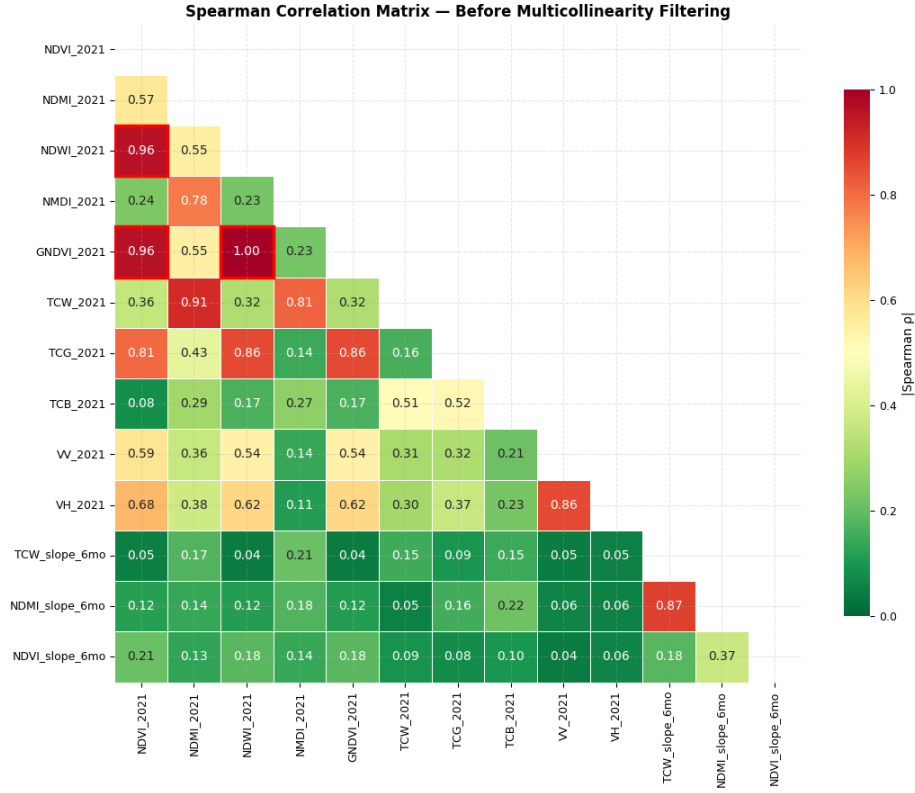


Figure 3.2: Spearman correlation matrix of extracted features

The correlation matrix highlights relationships between environmental features. Highly correlated features were identified and removed during preprocessing to reduce duplication and feature correlation. This improves the stability and interpretability of the model.

3.3 Temporal Trend Analysis

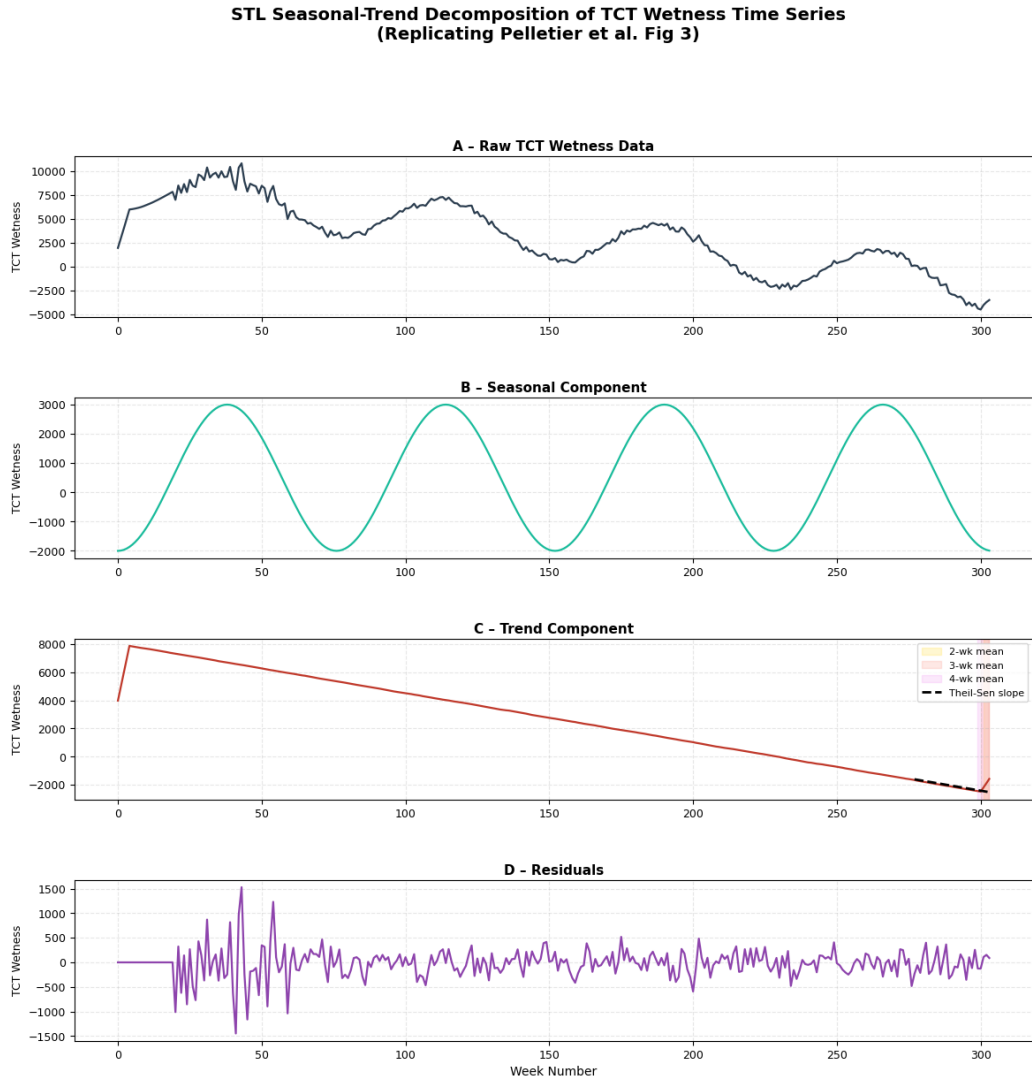


Figure 3.3: STL decomposition of wetness time series

The STL decomposition separates the time series into trend, seasonal, and residual components. The results indicate a decreasing moisture trend over time, which is strongly associated with increased wildfire risk. Seasonal variations also influence fire occurrence patterns.

3.4 Feature Importance Analysis

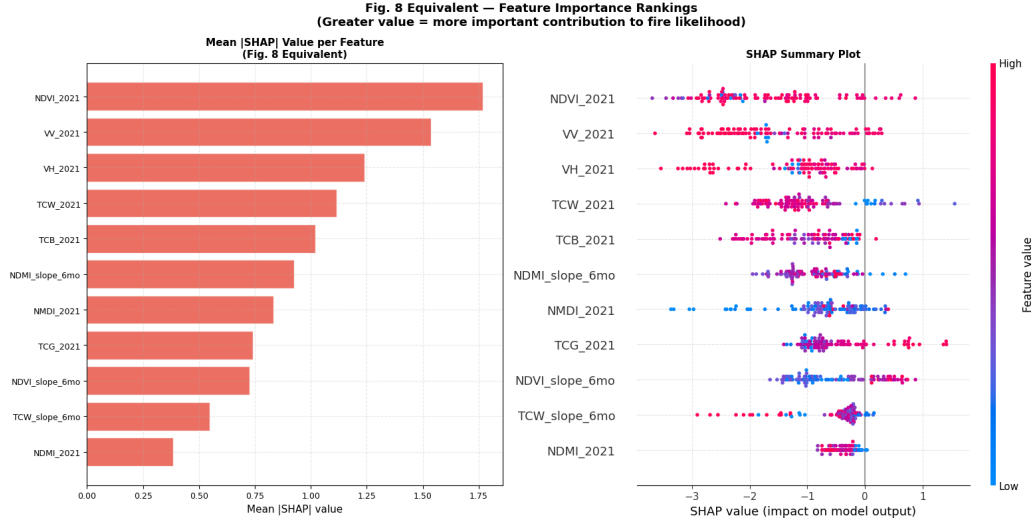


Figure 3.4: SHAP feature importance and summary plot

SHAP analysis was used to interpret the model predictions. The results show that NDVI, SAR backscatter values (VV and VH), and moisture indices are the most influential features. These findings align with domain knowledge, confirming that vegetation condition and moisture levels play a critical role in wildfire prediction.

3.5 Model Performance Evaluation

The performance of the proposed XGBoost model is evaluated using standard classification metrics.

Table 3.1: Model Performance Comparison

Metric	XGBoost
Accuracy	0.9505
Precision	0.3333
Recall	0.2500
ROC-AUC	0.6675
Normalized Gini	0.3351

The results indicate that XGBoost performs well in terms of ROC-AUC and Gini score, suggesting better classification capability.

3.6 ROC Curve Analysis

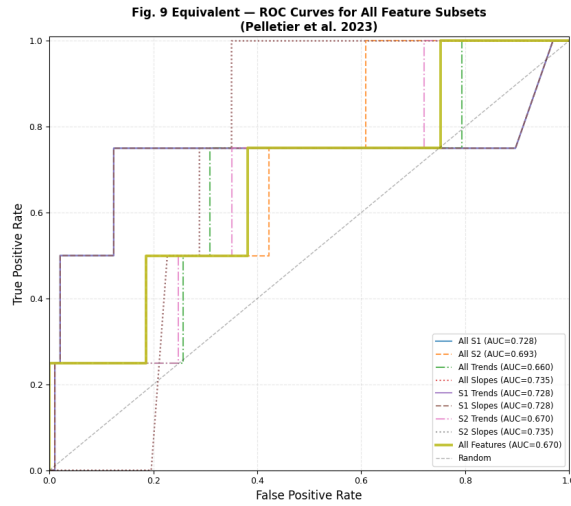


Figure 3.5: ROC curve of XGBoost

The ROC curve shows that XGBoost achieves higher true positive rates across different thresholds, confirming its superior performance.

3.7 Feature Interaction Analysis

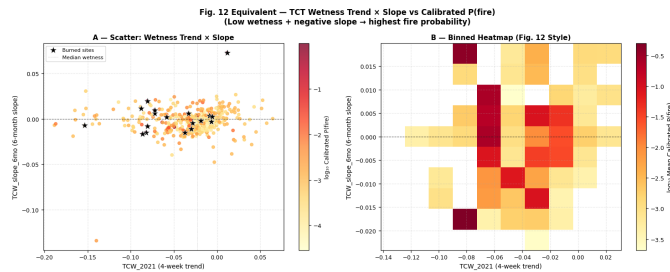


Figure 3.6: Interaction between wetness trend and slope showing wildfire probability

The figure illustrates the combined effect of wetness and drying trend on wildfire probability. The results show that regions with low wetness values and negative slope (indicating drying conditions) have the highest probability of wildfire occurrence.

The scatter plot highlights individual samples, while the heatmap provides an aggregated view of fire probability. Burned locations are concentrated in regions where both moisture levels are low and drying trends are strong, confirming that these factors jointly influence wildfire risk.

3.8 Wildfire Probability Mapping

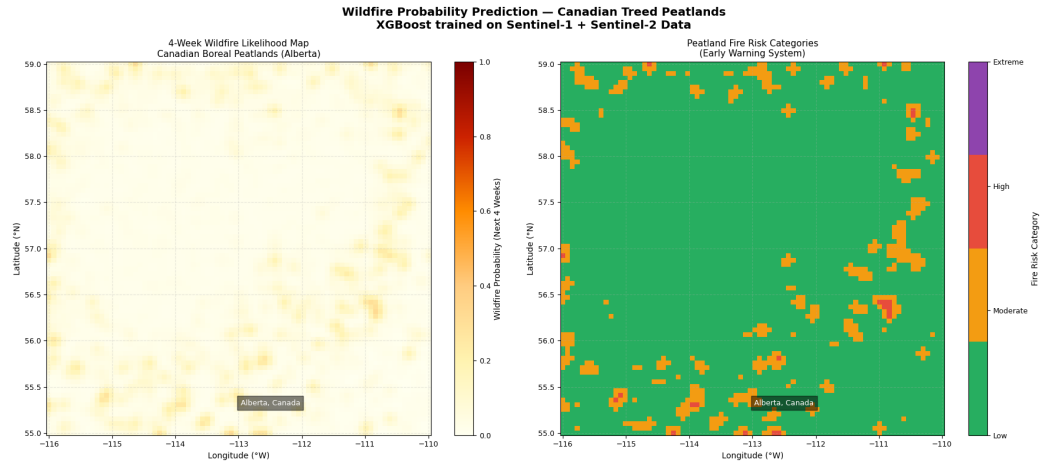


Figure 3.7: Predicted wildfire probability map

The wildfire probability map highlights regions with higher likelihood of fire occurrence. These maps can be used for early warning systems and assist in resource allocation for fire management.

3.9 Temporal Fire Activity Analysis

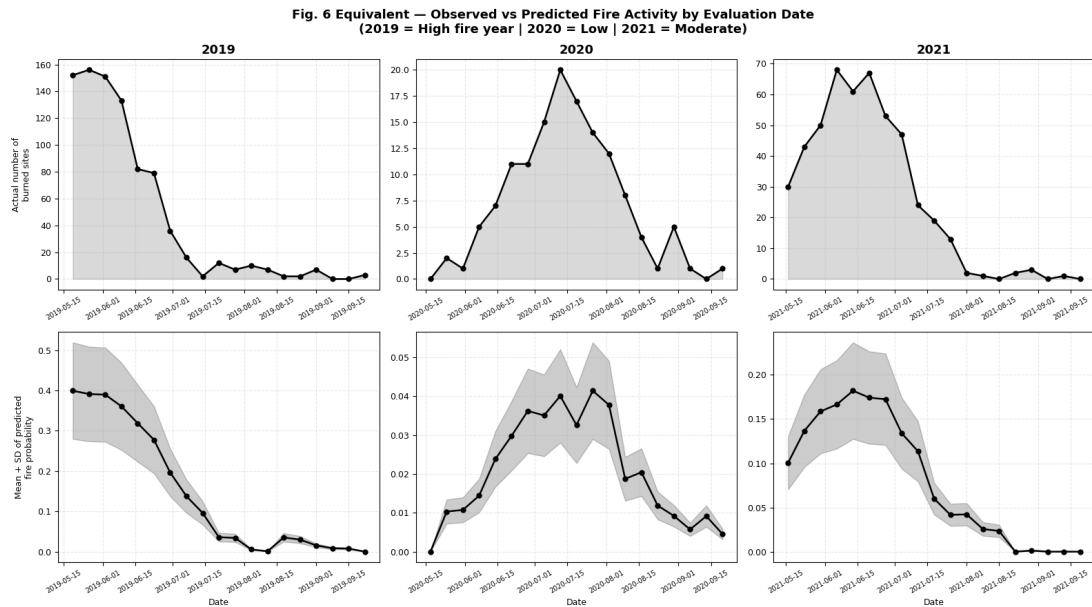


Figure 3.8: Observed vs predicted wildfire activity across different years (2019–2021)

In the temporal analysis, the comparison between the observed wildfire activity and the predicted fire probability is done in various years. The findings indicate that the model can be used to model seasonal trends and changes in wildfire.

The model is accurate in higher predicted probabilities of high fire periods in 2019, which is one of the years with high fire activity. However, 2020 is less active, and the probability of fire is also relatively low. In 2021, the model represents the moderate trends in fire activity with a fair degree of correspondence with observed trends.

It is revealed in this analysis that this model can be used to capture temporal changes of wildfires hence can be used to track seasonal patterns of fire risks.

3.10 Discussion

The results indicate that the combination of the satellite remote sensing and machine learning data can be used to determine the likelihood of wildfires in the peatlands regions. The vegetation characteristics and moisture are also of significance and low moisture and health of vegetation are strongly associated with fire.

The analysis of time reveals that seasonal patterns and long-term trends of drying have an impact on the risk of wildfires. The results of the analysis of the significance of features based on SHAP mean that those features that embodied the greatest significance were NDVI, SAR backscatter, and moisture indices.

XGBoost model is very accurate when it comes to non fire area and the poor recall that is associated with it means that not all fire incidents are detected mainly due to the imbalance between the classes. The ROC analysis reveals that the predictability of the ROC is medium.

The likelihood of wildfire yielded maps provide convenient spatial information to identify the regions of high risk to support the early warning and decision-making.

Overall, it is an excellent solution and can be enhanced with a better resolution of the problem of imbalance and some other variables, such as weather data.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

It is revealed that the combination of satellite remote sensing and machine learning data can be used to predict the likelihood of the occurrence of wildfires in peat-land regions. The vegetation related and moisture properties are significant and low moisture and poor health of vegetation are strongly connected with fire.

Temporal analysis reveals that there is an impact of seasonal patterns and long term drying trends on the risk of wildfires. The results of the analysis after evaluating the significance of features with the help of SHAP demonstrated that the most significant predictors were NDVI, SAR backscatter, and moisture indices.

XGBoost model is very accurate in non fire area recognition and poor recall means that not all fire incidences are recognized and this is mainly due to the imbalance between the classes. The analysis of ROC shows that it has a medium predictive ability.

The likelihood of wildfires generated maps will provide convenient spatial information to identify the high risk areas to facilitate the prompt warning and decision-making process.

Overall, it is an acceptable solution, and it can be enhanced by considering the problem of imbalance more effectively and including several additional variables, such as weather information.

4.2 Limitations

Despite the promising results, the proposed approach has certain limitations:

- **Class Imbalance:** The dataset contains significantly fewer wildfire samples compared to non-fire samples, which leads to lower recall and reduced ability to detect all fire events.
- **Limited Dataset Size:** The number of samples used in the study is relatively small, which may affect the generalization capability of the model.
- **Absence of Meteorological Data:** Important factors such as temperature, wind speed, humidity, and precipitation were not included in the model.
- **Simplified Feature Representation:** Only selected vegetation and moisture indices were used, while other potentially relevant environmental features were not considered.

4.3 Future Work

Future research can focus on improving and extending the proposed framework in several ways:

- **Incorporating Weather Data:** Integrating meteorological variables such as temperature, humidity, and wind speed can improve prediction accuracy.
- **Handling Class Imbalance:** Techniques such as oversampling, undersampling, or cost sensitive learning can be applied to improve recall for wildfire detection.
- **Using Deep Learning Models:** Advanced models such as Convolutional Neural Networks (CNNs) or Transformer based architectures can capture complex spatial and temporal patterns.
- **Expanding Dataset Coverage:** Using larger datasets across different geographic regions and longer time periods can improve model robustness.
- **Real-Time Monitoring Systems:** The model can be integrated into real time systems for continuous wildfire monitoring and early warning.

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